# AKSHAT NEEMA

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## Education

**PDPM IIITDM Jabalpur** — Jabalpur, Madhya Pradesh
*B.Tech in Computer Science and Engineering – GPA: 8.2/10.0* — *Expected May 2027*

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, System Design

## Experience

**Fusion ERP (Open-Source) | PDPM IIITDM Jabalpur** — Jabalpur, MP
*Hostel Management Module | Python, Django, PostgreSQL, Git, REST APIs* — *Aug 2025 – May 2026*

- Developed backend features for an institute-wide ERP system serving 1,000+ students, improving room allocation and hostel complaint management workflows.
- Automated room allocation and complaint tracking workflows, reducing manual administrative overhead by 40% and improving backend process reliability.
- Contributed to a production-oriented open-source codebase using Git, pull requests, code reviews, issue resolution, and collaborative software development practices.
- Developed and executed unit tests for critical backend logic, supporting technical validation and preventing regression defects across sprint cycles.

## Projects

**Expense Tracker** — *Apr 2026*
*TypeScript, React 18, Node.js, Express.js, REST APIs, SQLite, Vite*

- Developed a full-stack expense management application using React 18, TypeScript, Node.js, Express.js, REST APIs, and SQLite.
- Implemented API idempotency using a custom Idempotency-Key header to prevent duplicate submissions, with input validation and structured error handling.
- Built a responsive dashboard with transaction search, category and date-range filters, spending analytics, and recent activity; deployed the frontend and backend using Vercel and Render.

**Salon Booking Platform** — *Dec 2025*
*Flutter, Firebase, Firestore, Google Cloud Functions, REST APIs*

- Developed a cross-platform booking application supporting salon discovery, appointments, reviews, chat, cancellation, rescheduling, and salon-owner workflows.
- Implemented role-based access control and OTP authentication with Firebase for customer and salon-owner workflows.
- Designed Firestore data models and atomic transactions to handle concurrent bookings and real-time updates using streams and cloud functions.

**Coco – Audio-First AI Reading Companion** — *May 2026*
*Python, Gemini API, Google Vision API, OCR, OpenCV, TinyLlama, SHAP*

- Built an end-to-end AI application on Raspberry Pi 4B combining image acquisition, OCR, safety classification, LLM response generation, and text-to-speech for child interaction.
- Integrated Gemini 2.5 Flash Lite and Google Vision API with Python to process storybook images and generate contextual multilingual narration.
- Designed AI safety guardrails using TinyLlama intent classification and SHAP explainability to validate inputs before LLM generation.

## Technical Skills

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| **Languages** | JavaScript, TypeScript, Python, C++, Java, Kotlin, Dart, C |
| **Web & Backend** | Node.js, Express.js, REST APIs, React.js, TypeScript, JavaScript, Django, HTML, CSS |
| **Frameworks & Libraries** | Flutter, TensorFlow, Keras, PyTorch |
| **Databases** | MySQL, MongoDB, Firebase Firestore, PostgreSQL, SQLite |
| **DevOps & Cloud** | Git, GitHub, CI/CD, AWS, Google Cloud, Docker, VS Code, Android Studio, Jupyter Notebook |
| **AI & Data Processing** | Generative AI, LLM Integration, Gemini API, Google Vision API, Prompt Engineering, OCR, Computer Vision |
| **Core CS Concepts** | Software Engineering, OOP, DBMS, System Design, Operating Systems, Computer Networks, DSA |
| **Professional Skills** | Business-level English |

## Achievements & Certifications

- Selected for the Adobe GenSolve Hackathon, competing against and earning recognition among teams from premier engineering institutes across India. [Certificate]
- Completed the HP LIFE HP Power Lab Program in entrepreneurship and technology. [Certificate]
- Completed the SWAYAM Business Intelligence and Analytics course. [Certificate]